

Linear Regression

Chapter 3 – Introduction to Statistical Learning

Machine Learning Lecture

Motivating Example: Advertising Dataset

Goal: Predict product sales using advertising budgets.

Predictors:

- TV Advertising
- Radio Advertising
- Newspaper Advertising

Questions:

- ① Is advertising related to sales?
- ② Which media matter?
- ③ How large is the effect?
- ④ Can we predict future sales?
- ⑤ Is the relationship linear?
- ⑥ Are there interaction effects?

Simple Linear Regression

Model:

$$Y \approx \beta_0 + \beta_1 X$$

where

- Y = Response variable
- X = Predictor
- β_0 = Intercept
- β_1 = Slope

Prediction:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

Interpretation of Coefficients

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

Intercept

Expected value of Y when $X = 0$

Slope

Average increase in Y for one-unit increase in X

Least Squares Estimation

Residual:

$$e_i = y_i - \hat{y}_i$$

Residual Sum of Squares (RSS):

$$RSS = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Least Squares chooses coefficients that minimize RSS.

$$\min_{\beta_0, \beta_1} RSS$$

Least Squares Solution

Slope:

$$\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

Intercept:

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Key idea:

Best fitting line = Line with minimum RSS

Population vs Estimated Regression Line

True model:

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

Population regression line:

$$E[Y|X] = \beta_0 + \beta_1 X$$

Estimated line:

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$$

Estimated line changes from sample to sample.

Standard Errors

Coefficient estimates vary across samples.
Standard errors measure uncertainty.

$$SE(\hat{\beta}_1) = \sqrt{\frac{\sigma^2}{\sum (x_i - \bar{x})^2}}$$

Smaller standard error means:

- More precise estimate
- Narrower confidence intervals
- Stronger statistical evidence

Confidence Intervals

Approximate 95% CI:

$$\hat{\beta}_1 \pm 2SE(\hat{\beta}_1)$$

Interpretation:

If we repeatedly sample data, approximately 95% of intervals constructed this way will contain the true coefficient.

Hypothesis Testing

Test:

$$H_0 : \beta_1 = 0$$

vs

$$H_a : \beta_1 \neq 0$$

Test statistic:

$$t = \frac{\hat{\beta}_1}{SE(\hat{\beta}_1)}$$

Small p-value $\Rightarrow$ evidence of relationship between predictor and response.

Residual Standard Error (RSE)

Measures typical prediction error.

$$RSE = \sqrt{\frac{RSS}{n-2}}$$

Interpretation:

- Smaller RSE = better fit
- Same units as response variable

$$R^2 = 1 - \frac{RSS}{TSS}$$

where

$$TSS = \sum (y_i - \bar{y})^2$$

Interpretation:

- Proportion of variance explained
- $0 \leq R^2 \leq 1$

Examples:

- $R^2 = 0.90 \Rightarrow 90\%$ variance explained
- $R^2 = 0.20 \Rightarrow$ weak explanatory power

Multiple Linear Regression

When several predictors are available:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \varepsilon$$

Advertising example:

$$Sales = \beta_0 + \beta_1 TV + \beta_2 Radio + \beta_3 Newspaper + \varepsilon$$

Interpreting Coefficients

Important:

$$\beta_j$$

represents the effect of predictor X_j
while holding all other predictors fixed.

Example:

Radio coefficient = 0.189

Interpretation:

For fixed TV and Newspaper budgets, an additional \$1000 in Radio advertising increases sales by approximately 189 units.

Why Simple and Multiple Regression Differ

Example:

Simple Regression:

$$\text{Sales} \sim \text{Newspaper}$$

shows significant effect.

Multiple Regression:

$$\text{Sales} \sim \text{TV} + \text{Radio} + \text{Newspaper}$$

shows Newspaper is not significant.

Reason:

- Predictors can be correlated.
- Newspaper may act as a surrogate for Radio.

Overall Significance: F-Test

Null hypothesis:

$$H_0 : \beta_1 = \beta_2 = \dots = \beta_p = 0$$

Alternative:

At least one coefficient is non-zero.

$$F = \frac{(TSS - RSS)/p}{RSS/(n - p - 1)}$$

Large F-statistic $\Rightarrow$ evidence that predictors are useful.

Variable Selection

Goal:

Choose the most useful predictors.

Methods:

- 1 Forward Selection
- 2 Backward Selection
- 3 Mixed Selection

Evaluation criteria:

- Adjusted R^2
- AIC
- BIC
- Cross-validation

Confidence vs Prediction Intervals

Confidence Interval:

$$E[Y|X]$$

Average response.

Prediction Interval:

$$Y_{new}$$

Individual response.

Prediction intervals are always wider.

Qualitative Predictors

Categorical variables require dummy variables.

Example:

Gender:

$$Gender = \begin{cases} 1 & \text{Male} \\ 0 & \text{Female} \end{cases}$$

Regression:

$$Y = \beta_0 + \beta_1 Gender + \varepsilon$$

Coefficient measures difference between groups.

Interaction Effects

Sometimes predictors work together.

Model:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1 X_2 + \varepsilon$$

Interaction term:

$$X_1 X_2$$

captures synergy effects.

Example:

TV and Radio together may produce stronger sales than either alone.

Polynomial Regression

Linear models can capture non-linear patterns.

Quadratic model:

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \varepsilon$$

Still linear in coefficients.

Can fit using ordinary least squares.

Common Problems

- 1 Non-linearity
- 2 Correlated errors
- 3 Non-constant variance
- 4 Outliers
- 5 High leverage points
- 6 Collinearity

Diagnostics are essential before trusting results.

Residual Plots

Residual:

$$e_i = y_i - \hat{y}_i$$

Good residual plot:

- Random scatter
- No visible pattern

Bad residual plot:

- Curvature
- Funnel shape
- Clusters

Indicates model problems.

Outliers and Leverage

Outlier:

- Unusual response value

High Leverage:

- Unusual predictor value

Influential point:

- Changes fitted model substantially

Always investigate before removing.

Collinearity

Occurs when predictors are highly correlated.

Problems:

- Large standard errors
- Unstable coefficients
- Difficult interpretation

Detection:

- Correlation matrix
- Variance Inflation Factor (VIF)

Linear Regression vs KNN

Linear Regression:

- Parametric
- Low variance
- Highly interpretable

KNN:

- Non-parametric
- Flexible
- Can model complex relationships

When Does Each Method Win?

Linear Regression wins when:

- Relationship is approximately linear
- High dimensional data
- Interpretability matters

KNN wins when:

- Relationship is highly non-linear
- Large training dataset available
- Dimensionality is low

Key Takeaways

- 1 Linear Regression is the foundation of supervised learning.
- 2 Least Squares minimizes RSS.
- 3 Use t-tests and confidence intervals for inference.
- 4 Evaluate models using RSE and R^2 .
- 5 Multiple Regression controls for other predictors.
- 6 Interactions and polynomial terms extend flexibility.
- 7 Always check diagnostic plots.
- 8 Simpler models are often surprisingly powerful.

Questions?